Acid-Base Balance

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Acid–Base Balance

- The pH is term used for H^+ concentration
- Negative logarithm of H^+ concentration is taken for calculating the pH as given below.

$$pH = \log \frac{1}{H^+}$$

Acid–Base Balance

Normal pH : 7.38-7.42 , Average (7.4)

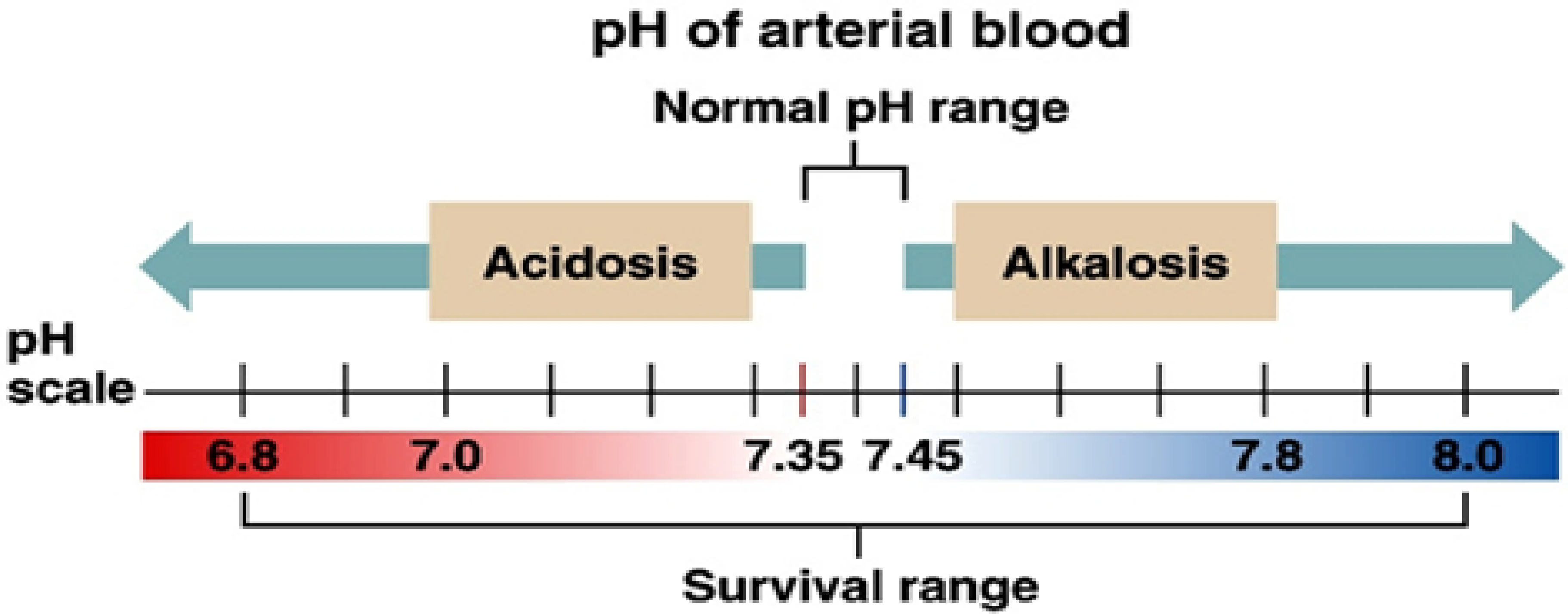
Acidosis

Physiological state resulting from abnormally low plasma pH (pH < 7.35= acedemia)

Alkalosis

Physiological state resulting from abnormally high plasma pH (pH > 7.45= alkalemia)

- pH < 6.8 or > 8.0 lead to death


pH of arterial blood
Normal pH range
Acidosis
Alkalosis
pH scale
6.8
7.0
7.35
7.45
7.8
8.0
Survival range

Control of pH

Mechanisms of control of pH;
#Buffer system
#Respiratory system
#The renal system

The respiratory system;

*Acts within seconds to minutes
The control acts by stimulation of 2 types of chemoreceptors .

Peripheral; stimulated by low Po_2- high Pco_2 and low pH in the plasma.

Central; stimulated by high Pco_2 and low pH in the CSF.

Stimulation by low pH;

correction of the low pH Results in stimulation of respiratory centre which causes hyperventilation that results in washout of CO_2 and

Stimulation of chemoreceptors by high pH;

High pH decreases stimulation of the chemoreceptors and this causes hypoventilation resulting in accumulation of CO_2 and correction of the high pH.

The renal system

Kidney plays an important role in maintenance of acid base balance by

-Reabsorption of HCO_3^-

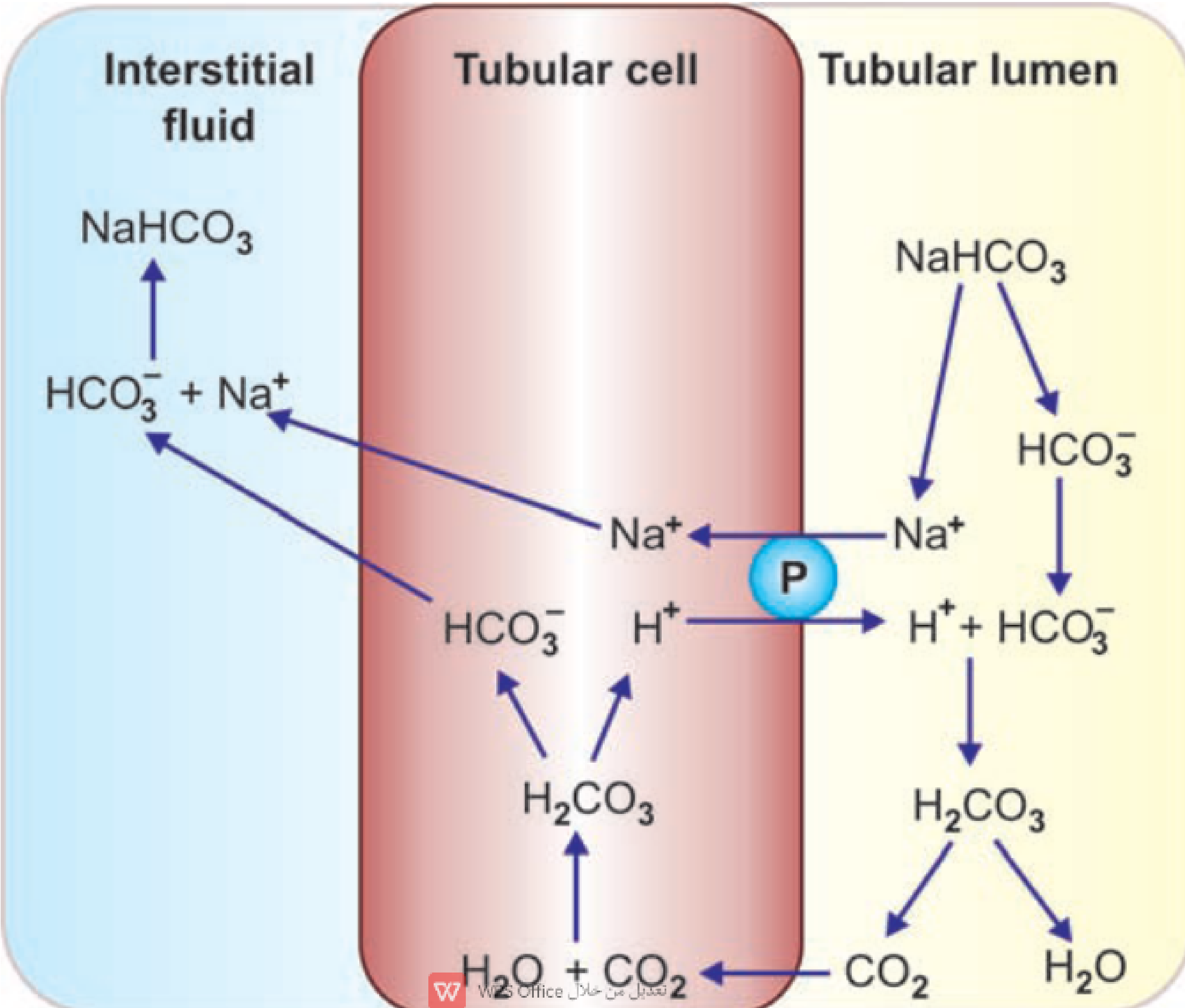
-Synthesis of HCO_3^-

-Secretion of H^+

Reabsorption of HCO_3^-

occur when the pH of urine is < 6.5 reabsorption occurs as follows;

$HCO_3^- + H^+ = H_2CO_3 = H_2O + CO_2$


Interstitial fluid
Tubular cell
Tubular lumen
$NaHCO_3$
$HCO_3^- + Na^+$
Na^+
HCO_3^-
H^+
H_2CO_3
$H_2O + CO_2$
P
$NaHCO_3$
HCO_3^-
Na^+
$H^+ + HCO_3^-$
H_2CO_3
CO_2
H_2O

- Then CO_2 diffuses inside PCT cells
- This reversible reaction between water and carbon dioxide proceeds at the presence of carbonic anhydrase enzyme.

CO_2 reacts with H2o at the presence of carbonic anhydrase enzyme results H^+ + HCO_3^-

H^+ is secreted to the lumen antiported with Na

HCO_3^- enters the interstitium co-transported with Na

- basolateral membrane At the collecting ducts HCO_3^- is exchanged with Cl^- through the

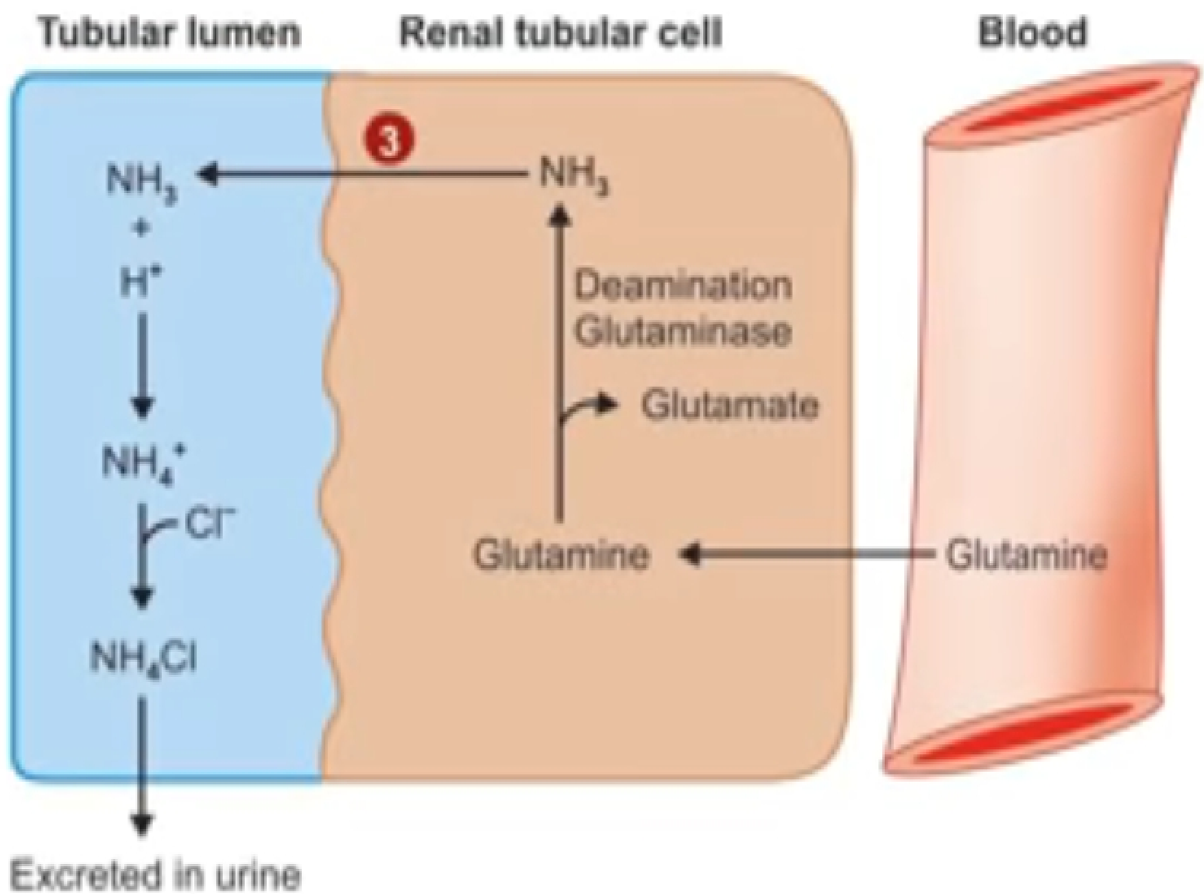
Synthesis of HCO_3^-;

Within PCTcells reaction between CO_2 (produced by metabolism within the cell)

and water formed bicarbonate.

Formation of amonia:

from amino acid glutamine during ammonia formation



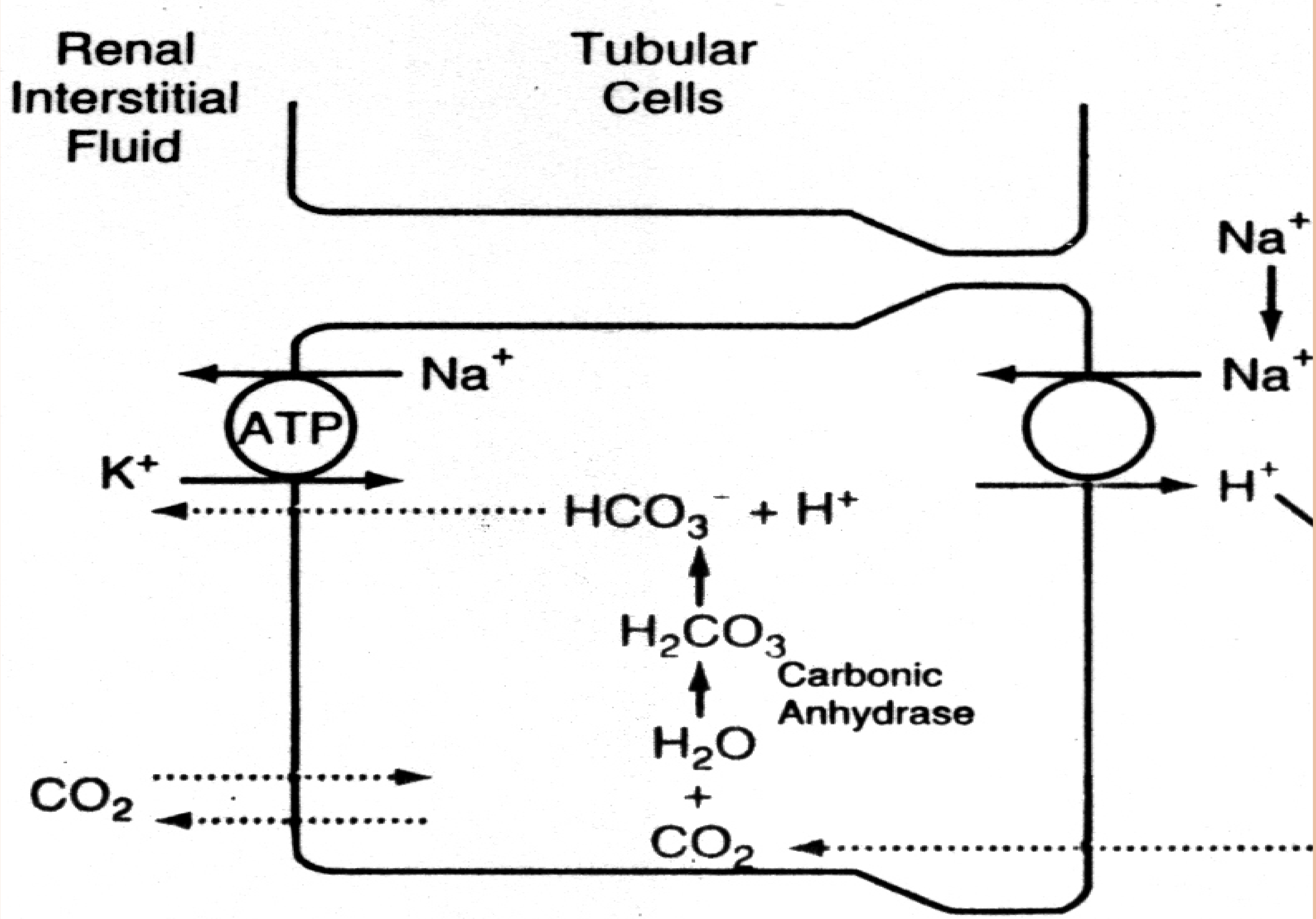
Formation of ammonia and excretion of ammonium ions in the urine.
Ref – Pankaja Naik, Biochemistry,5th Edition

Secretion of hydrogen;

proximal tubules.

collecting ducts

- In the PCT cells occurs by the following mechanisms;
- antiported with Na+
- H+ -ATPase pump
- In the collecting ducts occurs by ;
- H+ ATPase pump
- Antiported with K+


Renal Interstitial Fluid
Tubular Cells
Na+
Na+
Na+
ATP
K+
H+
HCO3- + H+
H2CO3
Carbonic Anhydrase
H2O
+
CO2
CO2

Acid base disturbances

- Metabolic acidosis due to low HCO_3^-
- Metabolic alkalosis due to high HCO_3^-
- Respiratory acidosis due to high CO_2
- Respiratory alkalosis due to low CO_2

Acidosis :

- Caused by decrease in the ratio of HCO_3 to H^+ in the renal tubular fluid.

Alkalosis:

- The ratio of HCO_3 to CO_2 in the extracellular fluid increases causing a rise in pH

- Respiratory acidosis and alkalosis require renal compensation
- Metabolic acidosis and alkalosis require renal and respiratory compensation

- Causes or respiratory acidosis;
- low pH and high PCO2

(Pco2 above 40 mm Hg due to hypoventilation)

- 1-respiratory centre depression by drugs or trauma
- 2-paralysis of respiratory muscles
- 3- diseases of the lung e.g emphysema

Compensation:

- Buffer system.

- Renal System
  1. Increases $[H^+]$ secretion
  2. Increases HCO_3 re-absorption and formation.

Causes of respiratory alkalosis

(high pH and lowPCO_2)

Pco_2 below 35 mmHg

1-hysteria (hyperventilation)

2- three-weeks residence at4000-m altitude.

3- - Hypoxic conditions

Compensation:

- Buffer system
- Renal system
- decreases $\{H^+\}$ secretion
- decreases HCO_3 re-absorption and formation.

Metabolic acidosis causes;

(low pH and low bicarbonate)
(1) failure of the kidneys to excrete metabolic acids normally formed in the body
(renal failure)
metabolic acids in the body e.g DKA and lactic acidosis (2) formation of excess quantities of
(3) addition of metabolic acids to the body by ingestion or infusion of acids
(4) loss of bases from the body fluids ; diarrhea .

Compensation:

- Buffer system
- Respiratory compensation:

 Increase in $\{H^+\}$ increases ventilation which lowers PCO_2 .

Renal Compensation:

-increasing H^+ secretion.

- increasing bicarbonate absorption and formation.

Metabolic alkalosis causes;

(high pH and high bicarbonate)

1- diuretics

All diuretics cause increased flow of fluid along the tubules, usually causing increased flow in the distal and collecting tubules. This leads to increased reabsorption of Na^+ from these parts of the nephrons

2-ingestion of alkali
such as sodium bicarbonate,
for the treatment of gastritis
or peptic ulcer.
3- loss of gastric acid e.g
vomiting -gastric suction.

◆ Renal Compensation:

- Decreases H^+ secretion.
- Decreases HCO_3 reabsorption and formation.

◆ Respiratory compensation

- Decreases ventilation to increase CO_2 concentration in the blood.

- The respiratory compensation;
- In acidosis -hyperventilation – lower PCO2
- In alkalosis – hypoventilation – higherPCO2

- The renal compensation;
- In acidosis there is increased;
- Reabsorption of bicarbonate
- Synthesis of bicarbonate
- Secretion of hydrogen ions
- Synthesis of ammonia
- In alkalosis all the above compensatory mechanisms are inhibited.

ANALYSIS OF DISORDERS

1-Measurement of Arterial Blood Gases

Pco_2 = 33 - 44 mmHg (40 mmHg)*
Po_2 = 75 - 105 mmHg (100 mmHg*)
pH = 7.35 - 7.45 (7.40 *)
[HCO_3^-] = 22 - 28 mEq/L (24 mEq/L*)
* Normal accepted value.

2- Measurement of Anion Gap

- The anion gap represents the difference between
  the concentration of the major plasma cation (Na^+) and the major plasma anions (Cl^- and HCO_3^-), and
- reflects the concentration of anions

- Anion Gap = ([Na+] +[K+])- ([Cl -] + [HCO3-])
- 8 - 16 mEq/ average =12 mmol L Under normal conditions the calculated anion gap is in the range of
- -the anion gap is used to differentiate between the causes of metabolic acidosis.

Causes of high anion gap;

1- DKA
2-severe exercise
3-alcohol toxicity
4- aspirin toxicity

Causes of normal anion gap

1- diarrhoea

2- renal tubular acidosis

causes of low anion gap

in multiple myeloma

3- the base excess

-normal range from -2 to +2 meq l

-A base excess of – 10 meq l indicates severe metabolic acidosis.

-A base excess of +10 meq l indicates severe metabolic alkalosis.


Thank